

每日一題39

單元1 數與式-雙重根號

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114翰林第=次模考#2

已知 $\sqrt{23+8\sqrt{7}} = a+b$ ，其中 a 為正整數， $0 < b < 1$ ，請問 $\frac{3}{a+3b} = ?$

- (1) $4+\sqrt{7}$
- (2) $8+2\sqrt{7}$
- (3) 7
- (4) $\sqrt{7}$
- ☒ (5) $\frac{\sqrt{7}}{7}$

<Sol>

$$\sqrt{23+8\sqrt{7}} = \sqrt{23+2\sqrt{112}} = \sqrt{16+\sqrt{7}} = 4+\sqrt{7}$$

$$\approx 4+2.\sim = 6.\sim$$

$$\therefore a=6, b=4+\sqrt{7}-6=\sqrt{7}-2$$

$$\frac{3}{a+3b} = \frac{3}{6+3(\sqrt{7}-2)} = \frac{3}{6+3\sqrt{7}-6} = \frac{3}{3\sqrt{7}} = \frac{1}{\sqrt{7}} = \frac{\sqrt{7}}{7} *$$

